

Problem Set 3

In this problem set, you will continue to be asked to work with the gamma distribution and you may find it useful to refer back to the previous problem set and/or solutions. I recommend Wikipedia as a reference, https://en.wikipedia.org/wiki/Gamma_distribution.

Keep your rendered `.pdf` to no more than 4 pages. Only provide code in the rendered `.pdf` when it is specifically requested.

1. Continue using four sets of values for the shape parameter α and rate parameter λ : $(\alpha_1, \lambda_1) = (0.5, 0.5)$, $(\alpha_2, \lambda_2) = (2, 1)$, $(\alpha_3, \lambda_3) = (0.5, 1)$, and $(\alpha_4, \lambda_4) = (2, 0.5)$, and store these values in a matrix `Y` with 4 rows and 2 columns, where the first column corresponds to the shape parameter, the second column corresponds to the rate parameter, and the rows correspond to the parameters in the order provided above.
 - (a) Consider a sequence of values for the x -axis of a plot given by `x <- seq(0.01, 6, by = 0.01)`. Create a matrix `A` with the same number of rows as `Y` and the same number of columns as the length of `x` with all elements equal to 0. Print the code you use to create this matrix to the rendered `.pdf`.
 - (b) Fill each row of `A` with the value of the gamma density evaluated at `x` for the parameter values in the same row of `Y`. Print the code you use to create this matrix to the rendered `.pdf`.
 - (c) Provide the maximum observed density value, i.e. the value of the largest element in `A`.
 - (d) Make a plot that is almost identical to the plots you made on Problem Set 2, with one change - make the limits of the y -axis 0 and the maximum density value you observe in `A`.
 - (e) Using `which.max` applied to each row of `A` and using `x`, create a vector of length 4 named `maxx` with elements that describe the value of `x` at which the density is largest for each set of parameter values. Print the code you use to create this vector to the rendered `.pdf`.
 - (f) Using the `ifelse` command and some simple mathematical operations applied to `Y` as well as what you know about the Gamma distribution, create a vector of length 4 named `modex` with elements that correspond to the mode of a Gamma distribution with the shape and rate parameters in each row of `A`. Print the code you use to create this vector to the rendered `.pdf`.

- (g) Summarize what you find in (e) and (f) using a 4×4 table made using `kable`, with a row for each pair of values named “Choice 1,” “Choice 2,” “Choice 3,” and “Choice 4” one column for the α values named “Shape,” one column for the λ values named “Rate,” one column for the corresponding mode named “Mode” and one column for the corresponding value of x where the density is maximized named “Approximate Mode”. Use at most 2 significant digits (you can adjust the number of digits printed by changing the argument `digits` that can be applied when using the `kable` function).
- (h) In at most two sentences, explain how the “Mode” and “Approximate Mode” columns compare and justify any differences you observe.
2. In class, we did a little exercise where we created our own data. Given a vector `x <- seq(1, 9, by = 1)`, we created a vector `y` by adding a random integer between -3 and 3 to each element of `x`, `y <- x + c(2, 2, -3, 0, -1, -2, 2, 3, 1)`.
- (a) Create a matrix with 9 rows and 100 columns called `Y` with each column defined as `x + sample((-3):3, 9, replace = TRUE)`. Print the code you use to create this matrix to the rendered `.pdf`.
- (b) Read the documentation of the function `sample` found by entering `help(sample)` into R and, given what you have learned about vectors and matrices, explain what `Y` is and how a row of `Y` relates to what we did in class in at most one sentence.
- (c) In class, we smoothed `y` by averaging observations with their nearest neighbors in each direction (`z` with `z[1] = (y[1] + y[2])/2`, `z[2] = (y[1] + y[2] + y[3])/3`, etc.) and then again by averaging observations with their k nearest neighbors in each direction (for $k = 2$, we have `w` with `w[1] = (y[1] + y[2] + y[3])/3`, `w[2] = (y[1] + y[2] + y[3] + y[4])/4`, `w[3] = (y[1] + y[2] + y[3] + y[4] + y[5])/5`, etc.). Create matrices `Z` and `W` of the same dimensions as `Y`, but with columns that contain smoothed columns of `Y` obtained by averaging observations with their nearest neighbors in each direction (for `Z`) and their nearest two neighbors in either direction (for `W`). You can use the `smoothing` function defined in `smoothing.R`. Print the code you use to create these matrices to the rendered `.pdf`.
- (d) Create vectors `ymean`, `zmean`, and `wmean` which contain the average of each row of `Y`, `Z`, and `W`, respectively. Print the code you use to create these vectors to the rendered `.pdf`.
- (e) Create vectors `ysd`, `zsd`, and `wsd` which contain the standard deviation of each row of `Y`, `Z`, and `W`, respectively. Print the code you use to create these vectors to the rendered `.pdf`.
- (f) Create a pair of plots by prefacing your `plot` commands with `par(mfrow = c(2, 1))` or `par(mfrow = c(1, 2))`. In the first plot, plot `x` against the row means you constructed in (d). Use points with `pch = 16`. Make sure to use a different color for set of row means and annotate them with a legend. In the second plot, plot `x` against the row standard

deviations you constructed in (e). Use points with `pch = 16`. Make sure to use a different color for set of row standard deviations and annotate them with a legend.